Chapter 17

Chest trauma

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Traumatic aortic dissection is described on 🕮 p452.

Principles of trauma management

Key facts

- Cardiothoracic surgeons are usually called to see trauma patients after initial evaluation and management has been carried out, but it is important to be aware of the general principles of trauma management.
- Initial approach consists of a focused ***primary survey*** aimed at detecting and treating immediately life-threatening conditions, followed by a more detailed secondary survey.
- Emergency department anterior thoracotomy (📖 p818) is reserved for loss of a pulse in the emergency department or en route. Survival depends on mechanism of injury (poor with blunt trauma and gunshots, better with stabbing).
- Chest trauma can be divided into blunt and penetrating: the majority of blunt chest trauma does not require operative intervention.
- Chest trauma is the third most common cause of trauma-related mortality behind head and spinal cord respectively.

Primary survey

This is aimed at identifying immediately life threatening injuries (see Box 17.1), and it consists of:

- A: Airway maintenance and cervical spine care.
- B: Breathing and ventilation.
- C: Circulation and control of hemorrhage.
- D: Disability and neurological state.
- E: Exposure.

Airway and maintenance of cervical spine

- Establish an airway: using basic life support maneuvers, simple Venturi mask, suction, nasopharyngeal tubes, oropharyngeal airways, endotracheal intubation, or rarely emergency cricothyroidotomy (📖 p248).
- The *cervical spine is immobilized, and all patient movements performed with in-line traction*, if the nature of the injury includes a significant impact or fall, until it is possible to exclude instability with a complete AP and lateral C-spine radiograph or CT.

Breathing and ventilation

Check ventilation by looking at chest expansion, ascultating for breath sounds, and monitoring O_2 saturations and blood gases where appropriate.

Box 17.1 Immediately life-threatening injuries

- Tension pneumothorax.
- Massive hemothorax.
- Open pneumothorax.
- Cardiac tamponade.
- Flail chest.

Causes of inadequate ventilation, all of which need immediate treatment are:
- Upper airway obstruction.
- Malplacement ET tube if used.
- Pneumothorax.
- Hemothorax.
- Massive lung contusion.
- Flail segment.

Circulation and control of hemorrhage

Signs of hypovolemic shock include confusion, pallor, tachycardia, tachypnea, oliguria. If peripheral access is difficult place a large lumen catheter into the femoral vein, or perform saphenous vein cut-down. Fluid resuscitation is aimed at expanding the intravascular space and may initially be titrated against pulse, BP, and later against urine output and central venous pressure. Causes of circulatory collapse include:
- Cardiac arrest.
- Hypovolemia: obvious bleeding site, or occult, e.g., fracture of femur or pelvis, intra-abdominal bleed, hemothorax.
- Cardiac tamponade.
- Cardiac contusion.
- Aortic transesction.

Disability and neurological state

Crucial in the evaluation of head and spinal injury, this is of less importance to the cardiothoracic surgeon than:

Exposure

Examine the patient front *and* back, swiftly but carefully for any evidence of penetrating or blunt trauma. Penetrating wounds below the fourth intercostal space may well have crossed the diaphragm, raising the specter of abdominal injury.

Secondary survey

This is a more detailed and complete clinical examination aimed at elucidating some form of basic history, and identifying and managing the following potentially life-threatening injuries:
- Rib fractures and flail chest.
- Pulmonary contusion.
- Simple pneumothorax.
- Blunt aortic injury.
- Blunt myocardial injury.

Investigations

First-line investigations performed once the primary survey and management is established include:
- CXR (C-spine and pelvis as indicated).
- TTE if pericardial effusion suspected.
- CT if there is any suspicion about injury to major thoracic structures, and the patient is hemodynamically stable.
- Aortography if aortic injury is suspected.

Emergency thoracotomy

Key facts

- Emergency thoracotomy is used to provide access to the chest for evaluation of the heart and descending aorta, and temporize catastrophic blood loss by cross-clamping the descending aorta, or alleviating cardiac tamponade.
- A ***clamshell*** provides rapid and excellent access to most areas of the chest except subclavian vessels, and does not require sternal saw: it carries significant morbidity in survivors but it is the incision of choice in the emergency room where the exact location of the problem may be unclear and rib retractors not immediately available.
- Indications include ***loss of cardiac output during transport or in the emergency department.***
- Knowledge of the causative agent, all entry and exit points is crucial in planning investigation and management of thoracic trauma.

Emergency thoracotomy in the emergency room

Where at all possible, every effort should be made to get the patient to an operating room where equipment, sterility, and personnel are readily available, approach can be more precisely tailored to the injury, and the patient resuscitated in a more controlled environment. If the patient has lost cardiac output clearly this is not possible.

Indications

- Cardiac arrest or imminent cardiac arrest after penetrating trauma.
- Limited role in cardiac arrest in setting of blunt trauma.

Incision

The approach is dictated by available expertise (do what you are able to do), and equipment (there may not be a sternal saw, or rib retractors), as much as by the nature of the injury. The options are:

- *Median sternotomy*: fast if a sternal saw and power source are available, excellent exposure to heart, both pleural cavities, hilums, but dependent on availability of sternal saw and suitable power source.
- *Anterior thoracotomy*: very fast, can be performed using just a scalpel, can be extended to improve what is very limited access to the pericardium and ipsilateral pleura. Difficult to carry out internal cardiac massage through this incision. Impossible to carry out definitive procedure if rib retractor not available.
- Bilateral extended anterior thoracotomy or *clamshell*: Fast, needs heavy scissors to divide sternum in addition to scalpel, excellent access to mediastinum and both pleurae. May have significant bleeding from IMAs bilaterally. Significant morbidity from wound in survivors.
- *Subxiphoid approach*: fast, but limited access to pericardium only.

Findings

Cardiac tamponade is the commonest finding, secondary to:

- RV lacerations more commonly than LV lacerations.
- Occasionally injury to coronary vessels, or great vessels.
- VSD is the commonest intracardiac injury.

- Occasionally there is no cardiac injury—the heart is empty from massive hemorrhage elsewhere: in this case internal cardiac massage is more effective than external chest compressions at getting resuscitation drugs to circulate while volume is replaced.

Emergency room clamshell

- Do not stop to place lines, or anesthetize or intubate the patient. Most of this can be done concurrently while you are opening chest, and once the patient has lost cardiac output time is critical.
- At a minimum you need gloves, a scalpel, a scissors and suction. Four clamps, a rib retractor, forceps and sutures are helpful.
- Stay calm, and communicate clearly with the person responsible for managing the airway, and the lead nurse who should be able to bring an emergency thoracotomy set and set up sterile suction with a Yankauer tip: there is usually an excited crowd of people trying to help perform a procedure most of them have never seen.
- Pour betadine on the patient's chest, put on sterile gloves (and gown if available), place drape over abdomen if available.
- Take scalpel and make incision following right to left inframammary crease across the midline, take it down to ribs and with care to intercostal muscles, protecting lung with forceps handle.
- Spend a minute to identify and clamp the proximal and distal ends of both internal mammary arteries: these retract and then bleed a lot once the patient gets a cardiac output.
- Divide the sternum with a heavy scissors or bone cutter.
- Place a retractor in if available, open the pericardium widely and carefully with a scissors.

Procedure

- Ventricular lacerations are repaired with pledgets of Teflon on 3/0 Prolene sutures, taking full thickness bites with great care to avoid iatrogenic trauma to the coronary arteries, particularly over the LV.
- Atrial and vessel lacerations are simply oversewn.
- Bleeding from the lung hilum is often best packed, especially if this is due to a deceleration injury.

Emergency thoracotomy in the operating room

- This is the optimal place to perform emergency surgery: if clinical condition permits the patient should be transferred immediately it becomes clear that thoracotomy is indicated.
- Do not delay transfer to insert additional monitoring lines.
- If you are dealing with an emergency from the invasive cardiology lab (e.g., tamponade after PCI, or catheter ablation) plan for a ***sternotomy*** and do not be persuaded to try a minimally invasive approach: these patients are sick, need rapid access, and may require a definitive procedure difficult via thoracotomy (repair of SVC, CABG, VAD, ECMO).
- Communicate clearly with the anesthesiologist so that they know your planned incision, timing, and findings.
- Generally, operating in a cardiac catheterization lab is even worse than trying to operate in the emergency room.

Blunt trauma

Key facts

- Emergency thoracotomy rarely indicated for management of blunt trauma: injuries may be diffuse and invariably unsuited to definitive repair or even temporizing in the emergency room.
- Diffuse bleeding is often best managed by aggressive transfusion and correction of all coagulopathy, including use of factor VII (📖 p146).
- Diaphragmatic injuries are described on 📖 p746 and p823.

Pulmonary contusions

- Generally caused by blunt trauma: characterized by alveolar and capillary hemorrhage of lung tissue resulting in parenchymal bruising, increased capillary permeability.
- May initially be asymptomatic, but often associated with hypoxia and significant ventilation/perfusion mismatch. ARDS (📖 p163).
- CXR demonstrates infiltrates, chest CT demonstrates areas of consolidation consistent with contusion.
- Therapy is primarily supportive. O_2, analgesia, physiotherapy. ET intubation and ventilation may become necessary. It is important to limit fluids, as the increased capillary permeability can lead to worsening and propagation of the contusion.

Tracheobronchial injury

- Tracheobronchial injuries can occur after both blunt and penetrating trauma. Blunt trauma or 'dashboard injury' is due to unrestrained motor vehicle passengers and sudden hyperextension of neck.
- The most common site of penetrating tracheal injuries is the cervical trachea, while the most common site of blunt tracheobronchial injuries is the right main stem bronchus, within 2.5cm of the carina, possibly due to an anatomic fixation point.
- Uncommon, approximately 0.5% of trauma cases, but may be life threatening and frequently associated with other major injuries.

Presentation

- Bronchial injuries may be asymptomatic, and have a delayed presentation months later with pneumothorax or bronchial stenosis.
- Dyspnea, respiratory distress, hoarseness, dysphonia. Surgical subcutaneous emphysema common. Pneumothorax in approximately half of cases and hemoptysis in a quarter.

Investigation

CXR often diagnostic. It may show disruption of the trachea or bronchial air column. Kumpe sign. CT useful in assessing the rest of the thorax. Flexible bronchoscopy diagnostic.

Management

- First step is airway control. Double lumen ET tubes should be avoided because their bulk and rigidity may exacerbate an injury. A single lumen tube should be placed beyond a tracheal injury, and lung isolation achieved by an Arndt bronchial blocker (📖 p647).

- For tracheal injuries where intubation is impossible secondary to the injury, a tracheostomy or cricothyroidotomy should be performed. Rarely, the patient can be intubated through the cervical injury.
- Small injuries may be managed conservatively, especially if the airway is intubated past the lesion.
- Proximal 2/3 of the trachea is accessed via a cervical collar incision.
- The distal half of the trachea, carina, proximal left main stem bronchus and entire right main stem bronchus can be accessed via a right posterolateral thoracotomy, the distal left main stem bronchus is accessed via a left posterolateral thoracotomy.
- Small lacerations can be repaired primarily with absorbable suture such as 4/0 Vicryl or 4/0 PDS.
- Larger bronchial injuries should be debrided and primary bronchial anastomosis similar to bronchoplastic sleeve anastomoses (📖 p688).
- Carinal injuries should be repaired, rather than resected.
- To relieve tension of the repair, different release techniques (📖 p755) can be used: (1) division of the pulmonary ligament, (2) suprahyoid laryngeal release may provide 1–2 cm of mobilization, (3) hilar release by dividing the pericardium around the inferior hilum.
- Associated organ injuries, e.g., esophageal wounds should be repaired.
- Viable tissue should be used to reinforce the repair and prevent fistulization, such as intercostal muscle, pleura patch, pericardial fat pad.
- A guardian chin stitch is used for tracheal and carinal repairs to prevent inadvertent tension (📖 p754).

Complications

- Anastomotic dehiscence (5%), or stricture.
- Fistulization (esophageal, arterial).
- Vocal cord paralysis.
- Over 75% of patients with tracheobronchial injuries secondary to blunt trauma die before arrival at the emergency department.

Chest wall, diaphragm, and pleura

Rib fracture:

Common. 4th–9th ribs most commonly fractured. Associated with underlying lung pleural, bronchial, and cardiac injuries. Lower rib fractures associated with renal, hepatic and splenic injuries. Fracture of the 1st rib (or sternum or scapula) indicates major trauma.

Management: adequate analgesia. May require regional nerve blocks. Epidural very effective anesthesia with minimal complications.

Flail segment:

- Flail chest is the fracture of three or more ribs in a row, with two or more fractures in each rib. The resulting chest wall 'island' exhibits paradoxical motion with respiration.
- Problems with flair chest often arise from potential for underlying pulmonary contusion: symptoms include dyspnea, hemoptysis, pain, respiratory splinting, respiratory distress.
- Usually obvious on clinical exam: can confirm with CXR and chest CT.
- Therapy is supportive. Mechanical ventilation if the patient is in distress, pain control with narcotics, PCA, rib blocks, or epidural analgesia.
- Care must be taken to limit fluids, as underlying pulmonary contusion is at risk for capillary leak.
- No clear data on the efficacy of early fixation. However, external plate fixation can be used to help wean off of mechanical ventilation.
- The flail segment will often fibrose with time, eliminating paradoxical movement with respiration.

Sternal fractures

- Sternal fractures are a marker of significant blunt trauma: associated with cardiac or great vessel injuries 18–62% of the time.
- Most common area of injury is the angle of Louis or the manubrial body junction.
- Symptoms include pain, dyspnea. Late presentations include chronic non-union with a sternal click, and late mediastinal abscess formation.
- Clinical diagnosis on examination, may be confirmed by CT chest—main value is CT angiography to look for vascular injuries.
- Most fractures require only supportive therapy, with pain control and aggressive pulmonary toilet.
- Unstable fractures may require debridement; with stabilization with external plates, or sternal wires. Abscesses require aggressive debridement, followed by closure with muscle or omental pedicled flap.

Clavicular fracture

- The majority, 80%, occur in the middle area of the clavicle, 15% laterally, and <5% medially. Lateral injuries occur from impact to the shoulder, and middle and medial injuries from blunt trauma to the chest.
- They present with well-localized pain, and CXR is diagnostic.
- Most fractures can be treated conservatively, with a sling and analgesia.
- Complex and/or open fractures may require debridement and fixation.

Diaphragmatic injury

- Blunt trauma, with acute intra-abdominal pressure can lead to laceration and disruption of the diaphragm.
- More common on the left side, as the liver serves as a barrier.
- Often asymptomatic. Symptoms include dyspnea, abdominal pain, atypical chest pain, dysphagia, bowel obstruction. May present late.
- Diagnosis can be difficult: CXR may demonstrate gastric bubble above the diaphragm. Chest CT with IV and oral contrast helpful.
- Diagnostic laparoscopy.
- Treatment is primary repair of the diaphragm. This can be done laparoscopically or thoracoscopically, or via a laparotomy or thoracotomy (📖 p747).

Pulmonary lacerations

Most common following penetrating chest trauma. Associated with hemothorax, pneumothorax, or hemopneumothorax. Extensive pulmonary lacerations seen following high-speed motor vehicle accidents. Often centrally located disrupting large vessels and major bronchi.

Management: peripheral lacerations may be oversewn or excised. They may heal following chest tube insertion. More extensive and central lacerations require surgical repair more urgently. Control of the hilum is required prior to any repair. Bronchial and vascular injuries are repaired. Laceration is usually left open with chest tube drainage.

Traumatic pneumothorax

Cause: direct lung injury from rib fracture or rapid deceleration. May be secondary to a rapid rise in intrathoracic pressure against a closed glottis causing rupture of the distal alveoli.

Presentation: dyspnea and chest pain. Examination reveals hyper-resonance to percussion and diminished or absent breath sounds on auscultation. A *tension pneumothorax* is life threatening and in addition may present with subcutaneous emphysema, mediastinal shift, acute respiratory distress, hemodynamic instability. Immediate chest decompression is performed by placing a wide-bore needle into the pleural cavity in the 2nd intercostal space in the midclavicular line.

Definitive management: chest tube insertion in the 4th or 5th intercostal space in the midaxillary line directing the chest tube anteriorly and superiorly. Thoracotomy rarely indicated.

Hemothorax

Causes: chest wall injury, lung parenchymal laceration, bronchial artery injury, intercostal vessel injury, major thoracic vascular injury. Chest radiograph diagnostic.

Treatment: tube thoracostomy (📖 p248) inserting a wide-bore, at least a 36F, tube in the 5th or 6th intercostal space in the midaxillary line directing the tube posteriorly. Bleeding ceases in 80–90% of cases.

Indications for surgery: immediate blood loss of >1L or bleeding continues in excess of 100mL/h for 4h or more. Blunt trauma to esophagus and thoracic duct rare.

Final thoughts

- Loupes: expensive but vital. Your program may buy them for you. Start with 2.5× extended field and practice suturing with them on at home. If you can buy a lightweight headlight that clips on to them as well, do it. If you wear contacts, keep a spare pair in your loupes case.
- Kit: Littman classic or cardiology stethoscope (engrave your name on it – they all look identical), this book, pen, cash, pager/phone.
- Food: ICU nurse station usually has the best list of numbers for food.
- Top five tips:
  - Treat every patient like you'd want to be treated.
  - Always have a plan B.
  - Treasure good mentors, hard work, luck, and common sense.
  - Things usually seem better after a good night's sleep.
  - But sleep is no substitute for experience.
- Top five books recommended for residency: This one for carrying around, and:
  - *Manual of Perioperative Care in Cardiac Surgery*, 5th edn., 2011. Robert M. Bojar MD. Blackwell Publishing. The gold standard for cardiac ICU care.
  - *Sabiston and Spencer. Surgery of the Chest*, 8th edn., 2009. Frank Selke, Pedro del Nido, and Scott Swanson. Two-volume good general cardiac, thoracic, and congenital surgery early reference book.
  - *Kaiser's Mastery of Cardiothoracic Surgery*, 2nd edn., 2006. Larry Kaiser, Irving Kron, and Thomas Spray. Lippincott Williams & Wilkins. Cardiac Surgery. Strong on operative technique with great editorial comments.
  - *Cardiac Surgery: Safeguards and Pitfalls in Operative Technique*, 4th edn., 2007. Siavosh Kohnsari and Colleen Flint Sintek. Lippincott Willliams & Wilkins. Best general cardiac surgical atlas there is, sadly does not cover general thoracic surgery.
- Top five online resources:
  - http://www.tsda.org. The best collection of education material online
  - http://www.ctsnet.org. Free access to text books, videos, editorials, jobs
  - http://www.myamericanheart.org. Full text of ACC/AHA guidelines
  - http://www.ncbi.nlm.gov. Pubmed search engine
  - http://www.hsforum.com. Lively online discussion group of CT surgeons.
- Top five courses for exams
  - The Utah Core Curriculum Review Course, http://www.corereview.org
  - The Birmingham Review Course, http://www.birminghamreviewcourse.co.uk
  - Applied Basic Science for Cardiothoracic Surgical Trainees, http://www.rcseng.ac.uk
  - European school for cardiothoracic surgery, http://http://school.eacts.org
  - SESATS self assessment, http://www.sesats.org